RF AMPLIFIER – LAB REPORT

Experiment: Frequency Response of an RF Amplifier

1. Aim

To study the frequency response of an RF amplifier and determine its power gain over the given frequency range.

2. Apparatus Required

RF Signal Generator, RF Amplifier, Spectrum Analyzer / Power Meter, RF Cables, Power Supply, Connectors and Terminations.

3. Theory

An RF (Radio Frequency) amplifier is an electronic circuit used to increase the power level of a radio-frequency signal while preserving its useful frequency characteristics.

The power gain of an amplifier is defined as the difference between output power and input power when both are expressed in dBm:

$G(dB) = P_{out}(dBm) - P_{in}(dBm)$

In this experiment, the input power is maintained at −10 dBm and the output power is measured at different frequencies. The frequency response shows how the gain changes with frequency.

4. Experimental Setup

RF Signal Generator → RF Amplifier → Spectrum Analyzer / Power Meter

The input power is set to −10 dBm and the frequency is varied over the required range.

5. Procedure

1. Connect the RF signal generator to the input of the RF amplifier.
2. Connect the output of the amplifier to the spectrum analyzer/power meter.
3. Switch ON the instruments and allow them to stabilize.
4. Set the input power to −10 dBm.
5. Set the required starting frequency.
6. Measure and record the output power.
7. Increase the frequency step by step and record the corresponding output power.
8. Calculate the gain using G = Pout − Pin.
9. Analyze the frequency response of the RF amplifier.

6. Observation Table

Frequency (MHz)	Pin (dBm)	Pout (dBm)	Gain (dB)
50	-10	13.81	23.81
150	-10	13.28	23.28
250	-10	12.92	22.92
350	-10	12.62	22.62
450	-10	12.52	22.52

Frequency (MHz)	Pin (dBm)	Pout (dBm)	Gain (dB)
550	-10	12.2	22.2
650	-10	12.23	22.23
750	-10	12.12	22.12
850	-10	11.87	21.87
950	-10	11.6	21.6
1050	-10	11.63	21.63
1150	-10	11.62	21.62
1250	-10	11.35	21.35
1350	-10	11.13	21.13
1450	-10	10.92	20.92
1550	-10	10.47	20.47
1650	-10	10.3	20.3
1750	-10	10.1	20.1
1850	-10	9.53	19.53
1950	-10	8.4	18.4
2000	-10	8.2	18.2

7. Sample Calculation

At 50 MHz:

Pin = –10 dBm

Pout = 13.81 dBm

Gain = Pout – **Pin = 13.81** – **(**–**10) = 23.81 dB**

8. Result

The frequency response of the RF amplifier was successfully studied. The maximum measured gain is **23.81 dB at 50 MHz** for an input power of –10 dBm. At 2000 MHz, the measured gain is **18.20 dB**.

9. Conclusion

The RF amplifier successfully amplifies the input RF signal. The measured gain is approximately 18–24 dB over the tested frequency range of 50 MHz to 2000 MHz. The gain gradually decreases with increasing frequency, indicating the frequency-dependent nature of the RF amplifier.

10. Precautions

1. All RF connections should be tight and properly terminated.

2. Do not exceed the rated input power of the amplifier.

3. Calibration of the measurement instruments should be checked before taking readings.

4. Avoid unnecessary movement or bending of RF cables during measurements.

5. Proper impedance matching should be maintained, generally 50 Ω in RF systems.